Numerical Analysis – Homework #2 (Solutions)

Utah State University Fall 2026

1.

Perform the following calculations using 3-digit rounding, 3-digit chopping and exactly:

(a) $12.3 + 0.0234$

> **Solution:** Exact: 12.3234. 3-digit rounding: 12.3. 3-digit chopping: 12.3.

(b) $-0.0321 + 0.000136$

> **Solution:** Exact: -0.031964. 3-digit rounding: -0.0320. 3-digit chopping: -0.0319.

(c) $12.3 - 0.0234$

> **Solution:** Exact: 12.2766. 3-digit rounding: 12.3. 3-digit chopping: 12.2.

2.

Write the polynomial

$$p(x) = 5x^6 + x^5 + 3x^4 + 3x^3 - x^2 + 1$$

in nested form.

> **Solution:** Coefficients (from x^6 to x^0): 5, 1, 3, 3, -1, 0, 1. Horner form:
>
> $$p(x) = \Big(\Big(\Big(\Big((5x+1)x+3\Big)x+3\Big)x-1\Big)x\Big)x+1.$$
>
> (Equivalently, show the $+0$ step explicitly if desired.)

3.

Suppose p^* must approximate p with relative error at most 10^{-3}. Find the largest interval in which p^* must lie for each value of p.

(a) $p = 150$

> **Solution:** $|p^* - p| \le 10^{-3}|p| = 0.15 \;\Rightarrow\; p^* \in [149.85,\, 150.15]$.

(b) $p = 1500$

Solution: $|p^* - p| \le 10^{-3}|p| = 1.5 \;\Rightarrow\; p^* \in [1498.5,\, 1501.5]$.

4.

Use 3-digit chopping (after each addition) for

$$\sum_{i=1}^{N} \frac{1}{i}.$$

(a) Find N such that N is the smallest integer for which your sum is not the exact solution (chopped to 3 digits).

Solution: Chop each term and each partial sum to 3 significant digits, and compare with the exact partial sum $H_N = \sum_{i=1}^{N} \frac{1}{i}$ chopped to 3 digits:

N	chopped running sum S_N	exact H_N	$\text{chop}_3(H_N)$
1	1.00	1	1.00
2	$\text{chop}(1.00 + 0.500) = 1.50$	1.5	1.50
3	$\text{chop}(1.50 + 0.333) = 1.83$	$11/6 \approx 1.8333$	1.83
4	$\text{chop}(1.83 + 0.250) = 2.08$	$25/12 \approx 2.0833$	2.08
5	$\text{chop}(2.08 + 0.200) = 2.28$	$137/60 \approx 2.2833$	2.28
6	$\text{chop}(2.28 + 0.166) = \mathbf{2.44}$	$49/20 = 2.45$	**2.45**

The smallest such N is $\boxed{N = 6}$, since $S_6 = 2.44 \neq 2.45 = \text{chop}_3(H_6)$.

Remark. If this is checked in MATLAB/Python with `fix(x*100)/100`, the computer finds $N = 7$ by mistake: in floating point $2.45 \cdot 100 = 244.999\ldots$, so $\text{chop}_3(H_6)$ comes out as 2.44. Exact (decimal) arithmetic gives $N = 6$.

(b) What is the absolute error for this value of N?

Solution: Using the exact $H_6 = \frac{49}{20} = 2.45$ as truth:

$$|E| = |2.45 - 2.44| = 0.01.$$

(c) What is the relative error for this value of N?

Solution:

$$\text{rel} = \frac{|E|}{|H_6|} = \frac{0.01}{2.45} = \frac{1}{245} \approx 4.0816 \times 10^{-3}.$$

5.

The number e can be calculated as

$$e = \sum_{n=0}^{\infty} \frac{1}{n!}.$$

Use three-digit chopping arithmetic to compute the following approximations to e. Also compute the relative and absolute errors. For the 'exact' value of e, use $e = 2.7182818$.

(a) $S_1 = \sum_{n=0}^{5} \frac{1}{n!}$

Solution: Evaluate with 3-digit chopping on each term and after each addition (order $n = 0 \to 5$): $S_1 = 2.70$.
Absolute error $|E| = |2.70 - 2.7182818| \approx 1.82818 \times 10^{-2}$.
Relative error $\approx 6.7255 \times 10^{-3}$.

(b) $S_2 = \sum_{n=0}^{5} \frac{1}{(5-n)!}$

Solution: This sums the same terms in reverse order $(1/5!, 1/4!, \ldots, 1/0!)$. With 3-digit chopping on terms and after each addition: $S_2 = 2.71$.
Absolute error $|E| = |2.71 - 2.7182818| \approx 8.2818 \times 10^{-3}$.
Relative error $\approx 3.0467 \times 10^{-3}$.
(Reversing the order reduces loss of significance.)